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Lab 3: Component Modelling & Architectural Pattern Selection

Architecture Selection – Peer Skill Exchange & Mentorship Network

Architecture Selection

We chose **Layered Architecture** for the **Peer Skill Exchange & Mentorship Network**.

Architectural Choice

The system is divided into **Presentation, Business, and Data layers**. The Presentation Layer contains the User Interface Component. The Business Layer contains the Order Manager Component, Skill & Mentorship Service Component, and Payment Service Component. The Data Layer contains the Database Component.

Reason 1

Layered architecture provides a clear separation of responsibilities. User interaction, mentorship and skill management, request processing, payment processing, and data storage are handled by separate components. This makes the system easier to understand, maintain, and modify.

Reason 2

The architecture is suitable for the Peer Skill Exchange & Mentorship Network because the system has several related functions that can communicate through well-defined interfaces. It provides a simple and organized structure without the additional operational complexity of a microservices architecture.

Security Advantage

The layered design allows sensitive operations such as user information and payment processing to be handled through business-layer components rather than allowing the User Interface to directly access the database. This provides better control over access to sensitive resources.

Performance Benefit

The separation of responsibilities allows frequently requested operations, such as searching for skills and mentors, to be handled by the Skill & Mentorship Service and supported by the database without placing unnecessary processing on the User Interface. This helps maintain efficient request processing.

Component Modelling Diagram

Layered Architecture

- Presentation Layer
- Business Layer
- Data Layer

